

NCR MANAGER · QUALITY MANAGEMENT SYSTEM

Data Model & Relationship Catalog

The 19 governed forms, business-ID numbering scheme and cross-record relationships that underpin the quality system.

DOCUMENT	VERSION	ISSUED	PLATFORM
DATA MODEL	1.0	June 27, 2026	Kinetic

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19 governed forms in kapp `ncrmanager`. Each record carries a human-readable **business ID** (the prefix scheme below) in addition to its Kinetic submission UUID. Cross-links are stored as business IDs (e.g. a CAPA's `Linked Complaint = "CMP-0001"`), so records reference authoritative IDs rather than copying data. Where historical integrity requires it (e.g. quantity affected, MRB decision) a snapshot value is stored on the record itself.

Numbering scheme (configurable)

PREFIX	RECORD	FORM SLUG	ID FIELD
QE	Quality Event	quality-events	Event ID
NC	Nonconformance	nonconformances	NC ID
CAPA	CAPA	capas	CAPA ID
CAPA-...-A#	CAPA Action	capa-actions	Action ID
CMP	Complaint	complaints	Complaint ID
AUD	Audit	audits	Audit ID
FND	Audit Finding	audit-findings	Finding ID
DOC	Document	documents	Document ID
CHG	Change Request	change-requests	Change ID
TRN	Training Record	training-records	Training ID
SUP	Supplier	suppliers	Supplier ID
SCAR	SCAR	scars	SCAR ID
RSK	Risk	risks	Risk ID
PRD	Product (DMR)	products	Product ID
EQP	Equipment	equipment	Equipment ID
SITE	Site	sites	Site ID
MR	Management Review	mgmt-reviews	Review ID
SIG	E-Signature	esignatures	Signature ID
AT	Audit Trail entry	audit-trail	Entry ID

Module groups

- **Master data:** sites, products, suppliers, equipment
- **Intake & investigation:** quality-events, nonconformances, capas, capa-actions, complaints
- **Audit:** audits, audit-findings
- **Document control:** documents, change-requests
- **People:** training-records
- **Supplier quality:** scars (+ suppliers)
- **Risk:** risks
- **Governance:** mgmt-reviews
- **Compliance substrate (append-only):** audit-trail, esignatures

Relationship graph (link fields)

Arrows point from the record that *holds* the reference to the record it references. The server's record-360 endpoint resolves these **forward** and scans the same fields **in reverse** to build the full neighborhood for any record.

```
complaints —Linked Risk→ risks
complaints —Linked Event→ quality-events
complaints —Linked CAPA→ capas
complaints —Duplicate Of→ complaints
quality-events —Linked NC→ nonconformances
quality-events —Linked CAPA→ capas
quality-events —Linked Complaint→ complaints
quality-events —Product / Supplier→ products / suppliers
nonconformances —Source Event→ quality-events
nonconformances —Linked CAPA / Linked Complaint→ capas / complaints
capas —Source Event→ quality-events
capas —Linked NC / Complaint / Risk / Change / SCAR→ ...
capa-actions —CAPA ID→ capas
audit-findings —Audit ID→ audits
audit-findings —Linked CAPA→ capas
audit-findings —Supplier→ suppliers
change-requests —Linked CAPA / Linked Product / Document→ ...
training-records —Linked CAPA / Linked Change / Document→ ...
scars —Supplier / Source Event / Linked CAPA→ ...
risks —Product / Linked CAPA / Linked Complaint→ ...
audits —Supplier / Site→ ...
audit-trail —Record ID→ (any record)
esignatures —Record ID→ (any record)
```

| Indexing rules applied

- Every field appearing in a KQL filter (Status, Type, Owner, Site, due dates, link/ID fields) has a **single** index; common pair-filters (e.g. `Status + Owner` , `Status + Site` , `Status + Risk Level`) have **compound** indexes. See `indexes` in `app.json` .
- Append-only forms are indexed on `Record Type` + `Record ID` (compound) so a record's full history and signature manifest are a single selective query.

| Data-integrity model

- **No silent overwrite of regulated records.** Edits to records in the *regulated set* (`quality-events` , `nonconformances` , `capas` , `complaints` , `audit-findings` , `change-requests` , `scars` , `risks`) require a **reason for change**; each changed field is written to `audit-trail` with old → new values, actor, timestamp, workflow state and correlation ID.
- **Append-only history.** `audit-trail` and `esignatures` are only ever inserted into, never updated.
- **Snapshots vs references.** Lots, quantities and decisions that must remain historically faithful are snapshotted onto the record; everything else is an authoritative ID reference.